

Attention

NLP: Fall 2025

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Machine Translation

$$\arg \max_{\mathbf{y}} p(\mathbf{y} \mid \mathbf{x})$$

Target language

Source language

Neural Machine Translation

$$\arg \max_{\mathbf{y}} p(\mathbf{y} \mid \mathbf{x}) = \prod_{t=1}^T p(y_t \mid \{y_1, \dots, y_{t-1}\}, c),$$

$$\mathbf{y} = (y_1, \dots, y_{T_y}).$$

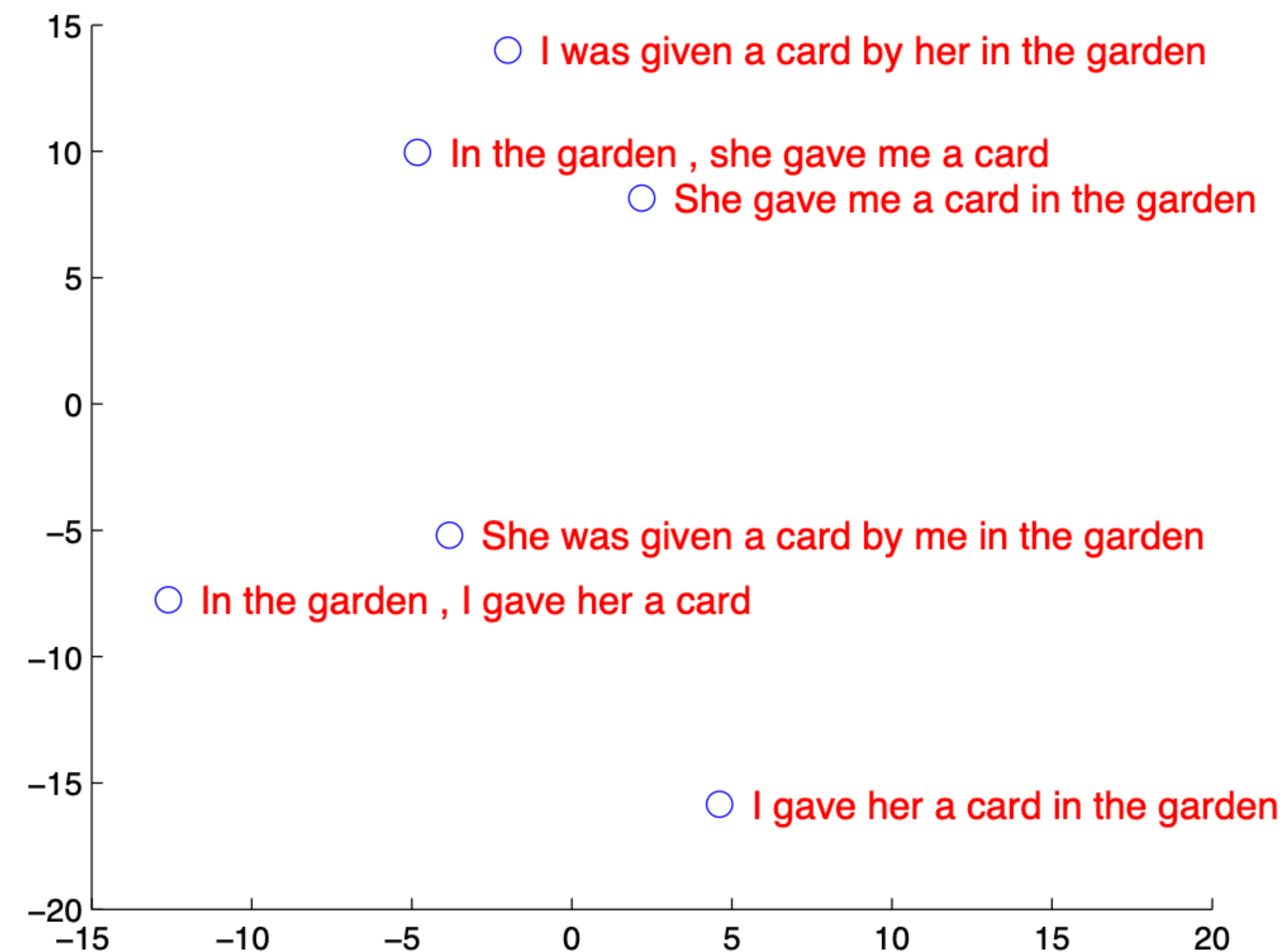
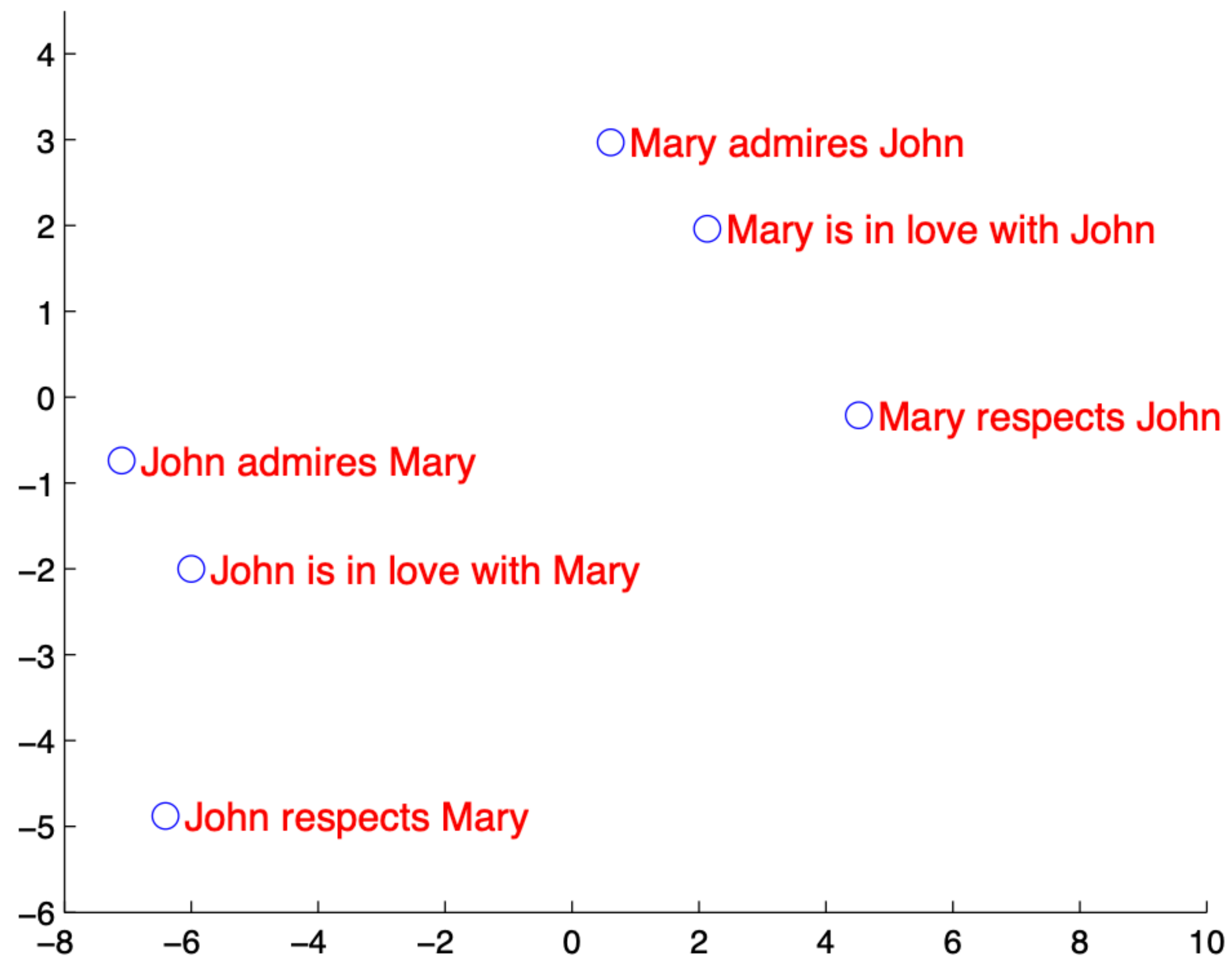
$$c = q(\{h_1, \dots, h_{T_x}\})$$

$$\mathbf{x} = (x_1, \dots, x_{T_x}).$$

h_i is a vector representation of x_i

c is a vector assembled from all the h_i vectors

s_i is a vector representation of y_i



Sequence to Sequence Learning with Neural Networks <https://arxiv.org/abs/1409.3215>

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

Dzmitry Bahdanau

Jacobs University Bremen, Germany

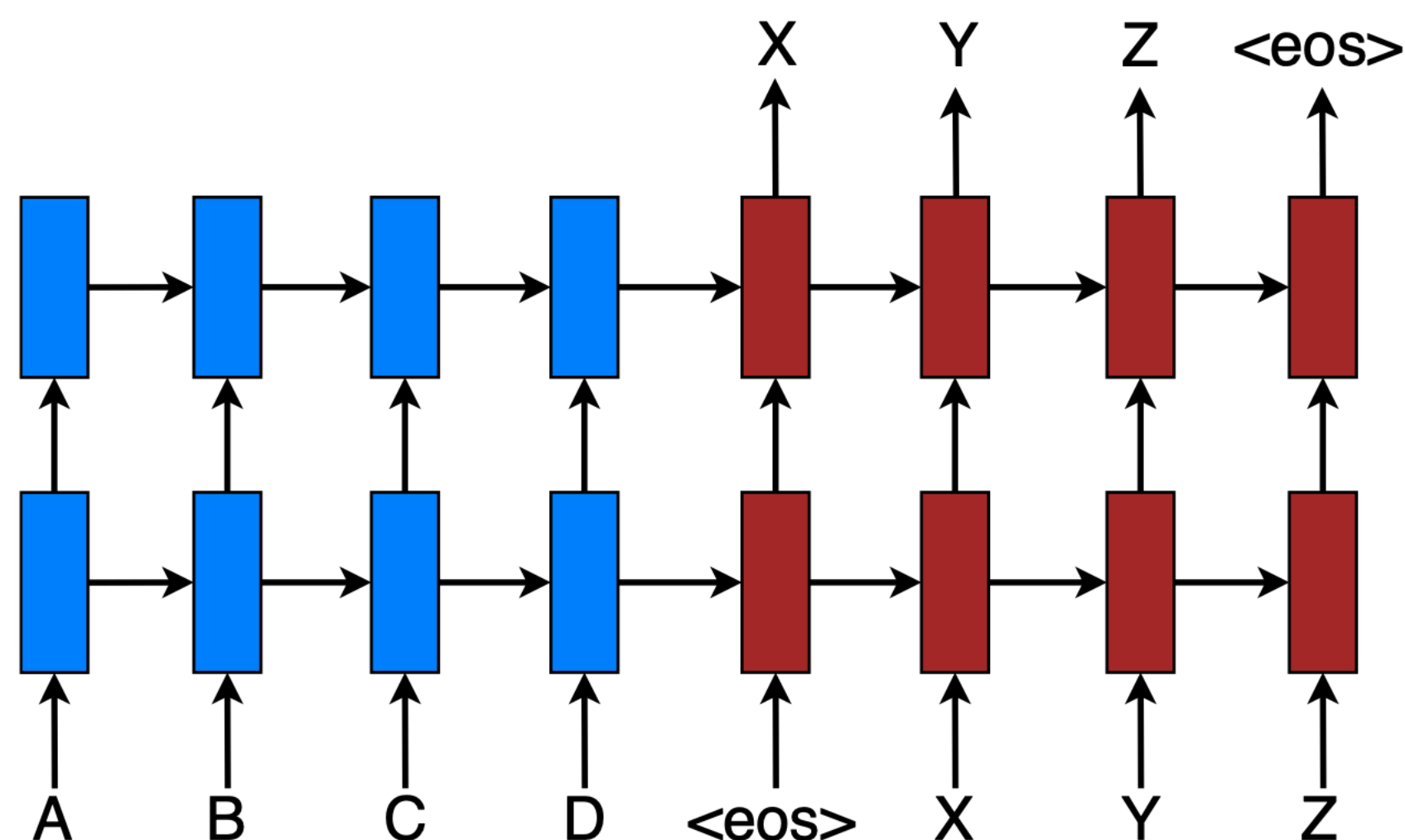
KyungHyun Cho Yoshua Bengio*

Université de Montréal

ICLR 2015

<https://arxiv.org/abs/1409.0473>

Effective Approaches to Attention-based Neural Machine Translation

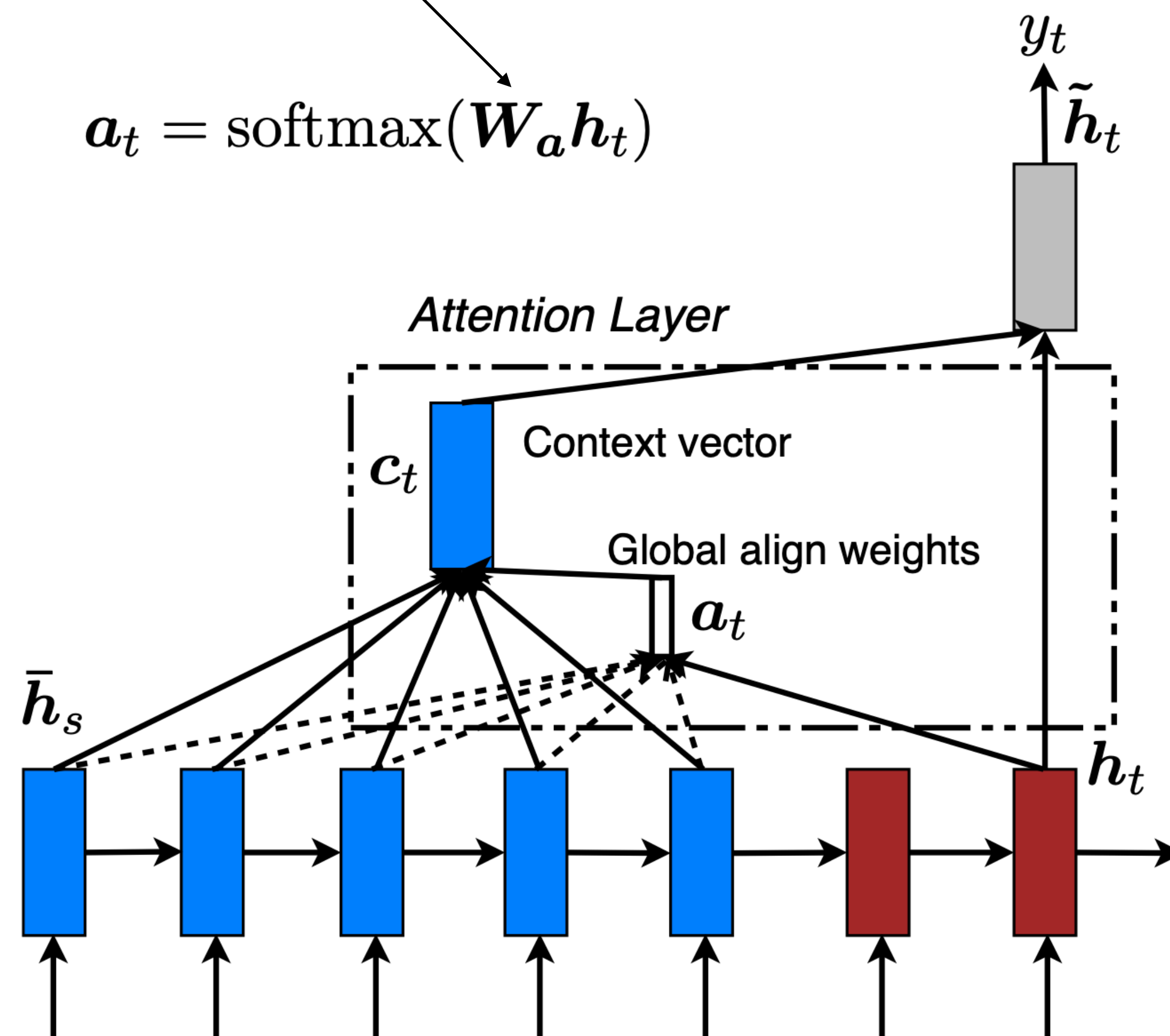


$$\text{score}(\mathbf{h}_t, \bar{\mathbf{h}}_s) = \begin{cases} \mathbf{h}_t^\top \bar{\mathbf{h}}_s & \text{dot} \\ \mathbf{h}_t^\top \mathbf{W}_a \bar{\mathbf{h}}_s & \text{general} \\ \mathbf{v}_a^\top \tanh(\mathbf{W}_a [\mathbf{h}_t; \bar{\mathbf{h}}_s]) & \text{concat} \end{cases}$$

<https://arxiv.org/abs/1508.04025>

$$\frac{\exp(\text{score}(\mathbf{h}_t, \bar{\mathbf{h}}_s))}{\sum_{s'} \exp(\text{score}(\mathbf{h}_t, \bar{\mathbf{h}}_{s'}))}$$

$$\mathbf{a}_t = \text{softmax}(\mathbf{W}_a \mathbf{h}_t)$$



Attention Networks in Neural Machine Translation

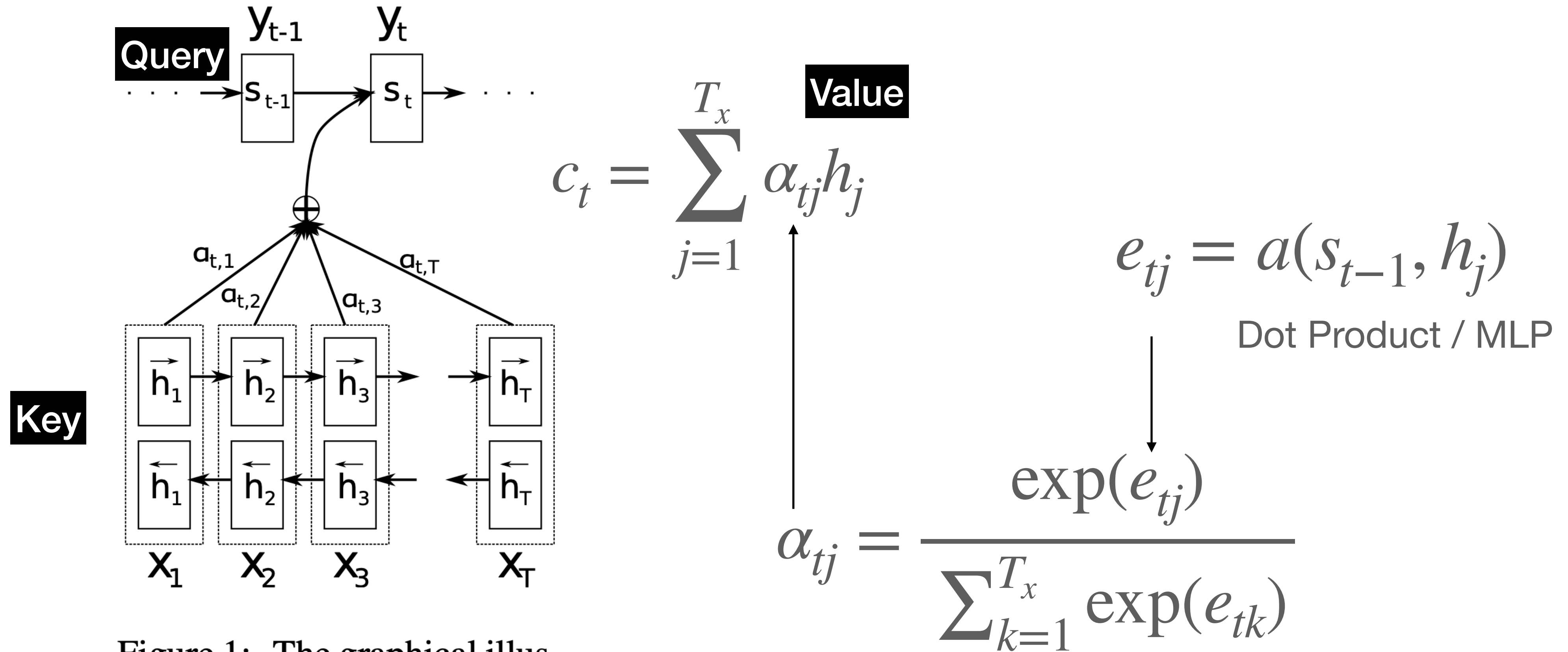
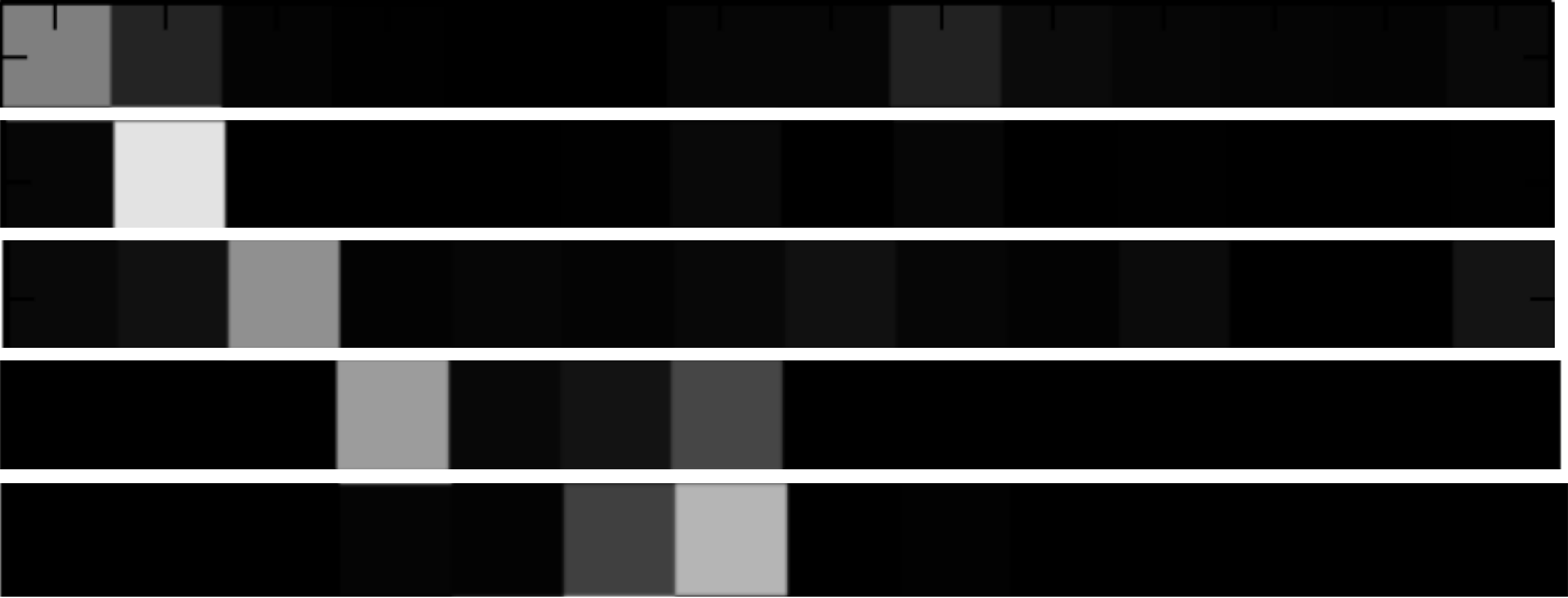


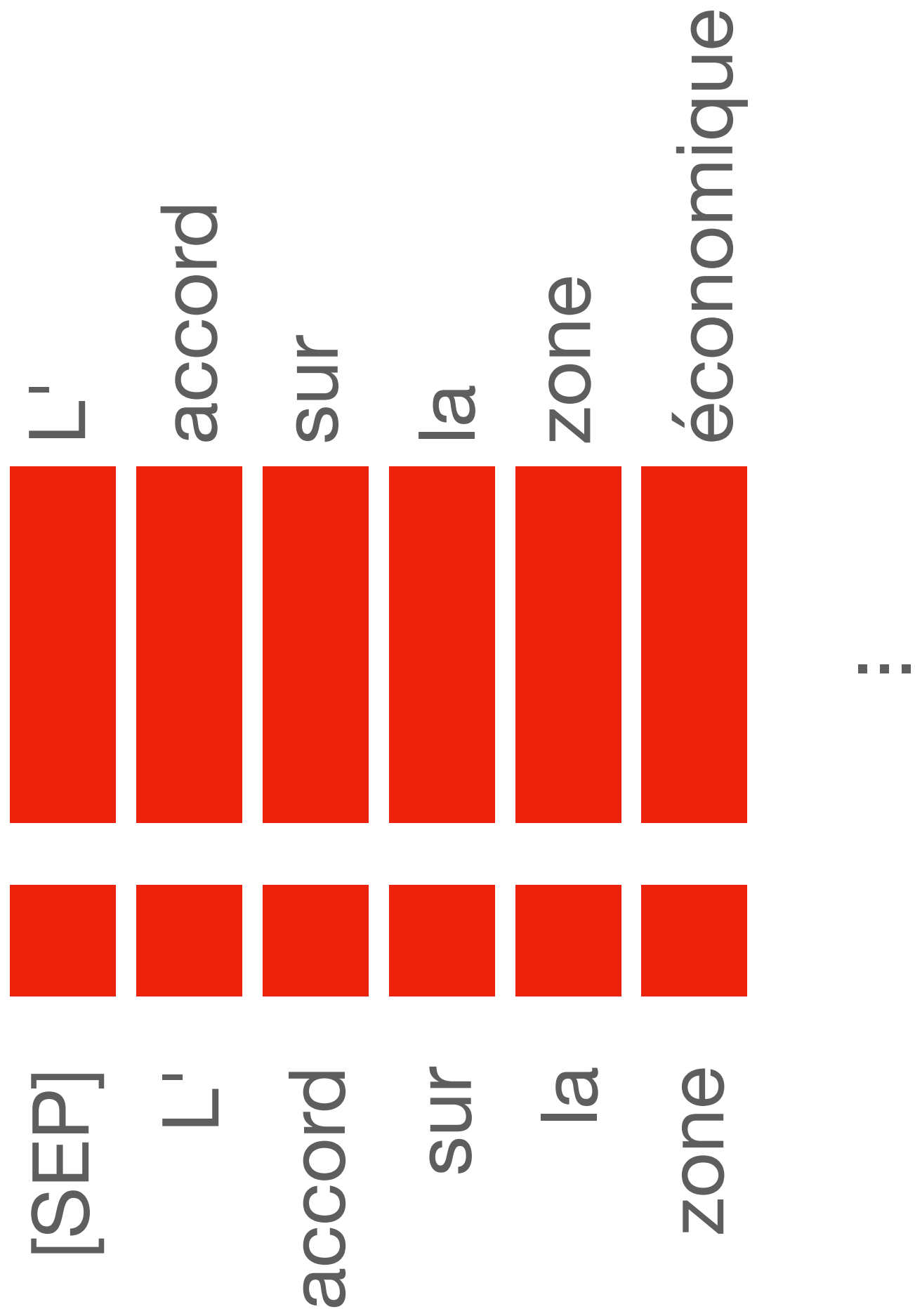
Figure 1: The graphical illustration of the proposed model trying to generate the t -th target word y_t given a source sentence $(x_1, x_2, \dots, x_T)$.



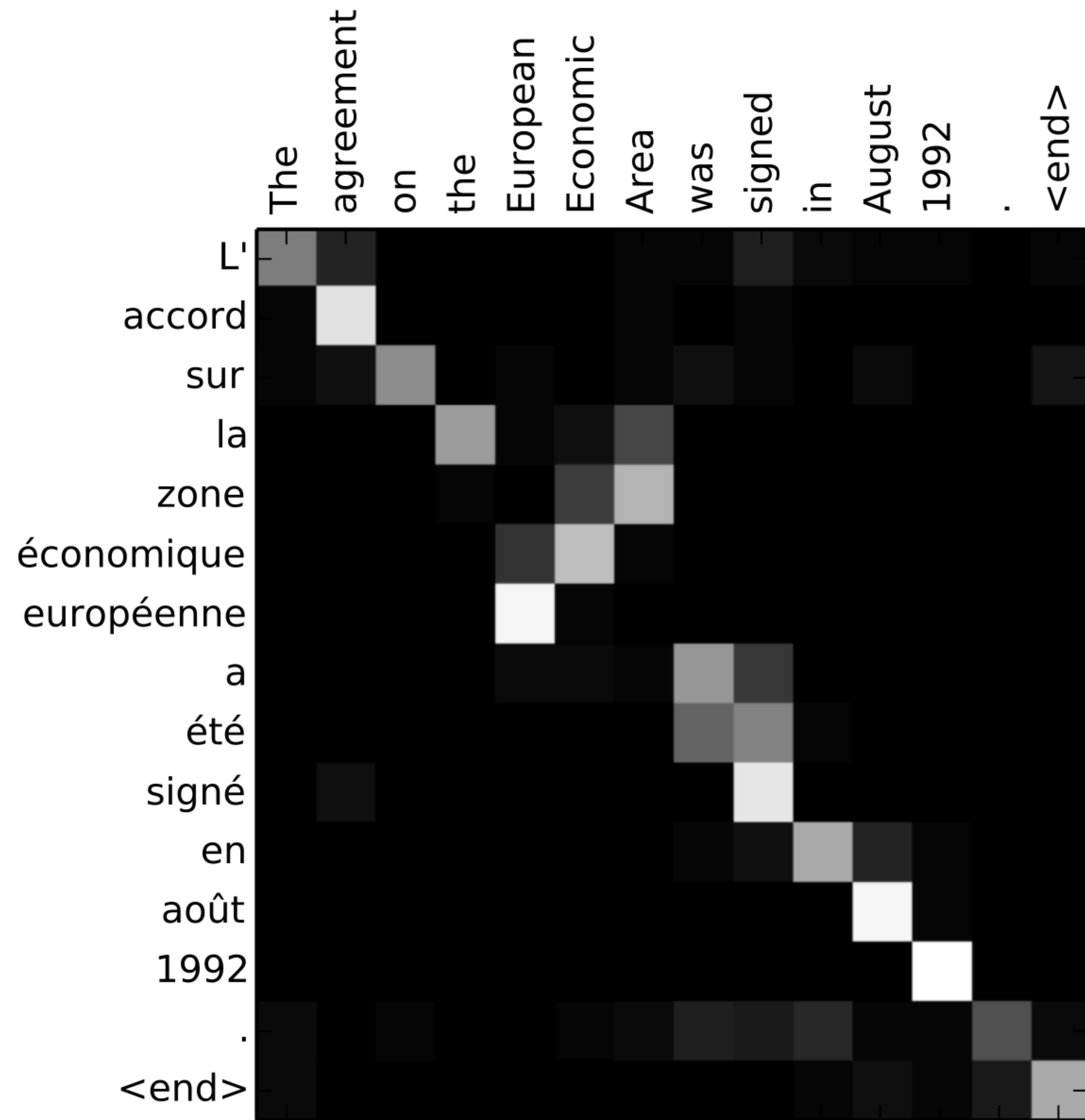
L'
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The
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August
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[SEP]
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Source

An admitting privilege is the right of a doctor to admit a patient to a hospital or a medical centre to carry out a diagnosis or a procedure, based on his status as a health care worker at a hospital.

No attention

Un privilège d'admission est le droit d'un médecin de reconnaître un patient à l'hôpital ou un centre médical d'un diagnostic ou de prendre un diagnostic en fonction de son état de santé.

With attention

Un privilège d'admission est le droit d'un médecin d'admettre un patient à un hôpital ou un centre médical pour effectuer un diagnostic ou une procédure, selon son statut de travailleur des soins de santé à l'hôpital.

Source

This kind of experience is part of Disney's efforts to "extend the lifetime of its series and build new relationships with audiences via digital platforms that are becoming ever more important," he added.

No attention

Ce type d'expérience fait partie des initiatives du Disney pour "prolonger la durée de vie de ses nouvelles et de développer des liens avec les lecteurs numériques qui deviennent plus complexes.

With attention

Ce genre d'expérience fait partie des efforts de Disney pour "prolonger la durée de vie de ses séries et créer de nouvelles relations avec des publics via des plateformes numériques de plus en plus importantes", a-t-il ajouté.